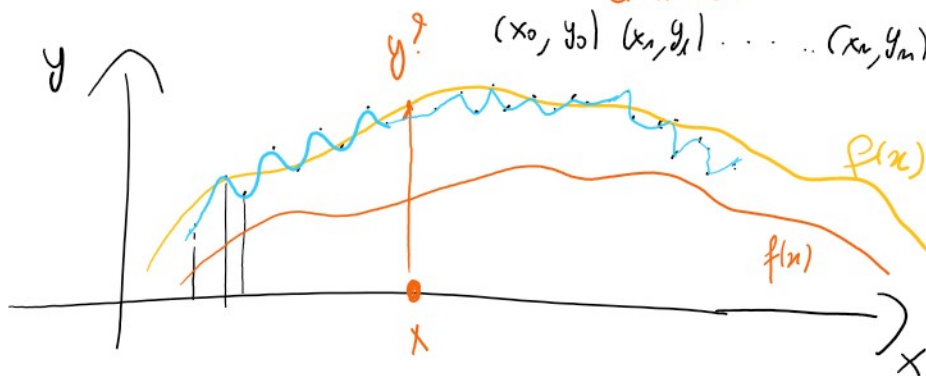


14/5/2025

mercoledì 14 maggio 2025 11:19

APPROSSIMAZIONE DI DATI



APPROSSIMAZIONE POLINOMIALE $\rightarrow p(x) = a_0 + a_1x + a_2x^2 + \dots + a_mx^m$

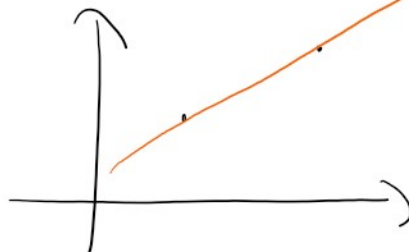
INTERPOLAZIONE f tale che $f(x_i) = y_i \quad i = 0, \dots, n$

MINIMI QUADRATI f tale che $\sum_{i=0}^n (f(x_i) - y_i)^2$ sia MINIMA

FISSATO n GRADO del polinomio

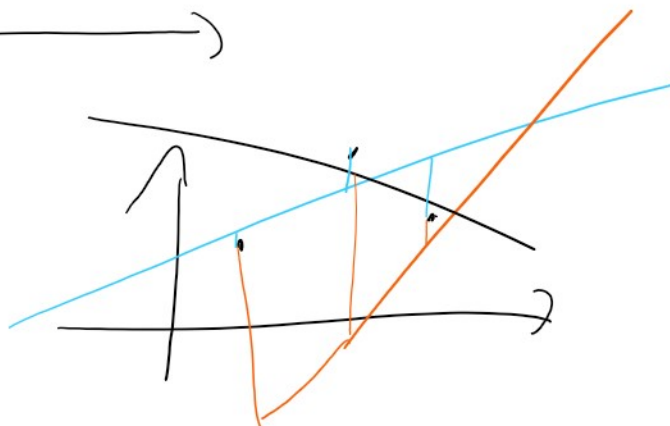
$n=1$

$(x_0, y_0) \quad (x_1, y_1)$



$n=2$

$(x_0, y_0) \quad (x_1, y_1) \quad (x_2, y_2)$



INTERPOLAZIONE

$(x_0, y_0) \cdot (x_1, y_1) \cdot \dots \cdot (x_n, y_n) \quad n+1$ punti

$\Rightarrow$ devo considerare un polinomio di grado n

devo considerare un polinomio di grado n

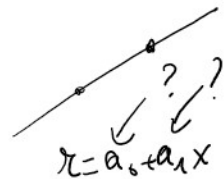
INTERPOLAZIONE

$$P_m(x) = a_0 + a_1x + \dots + a_mx^m$$

$(x_0, y_0) \quad (x_1, y_1) \quad (x_2, y_2) \quad \dots \quad (x_m, y_m) \quad m+1 \text{ punti}$

$$P_m(x_i) = y_i \quad i = 0 \dots m$$

$a_0 = ? \quad a_1 = ? \quad \dots \quad a_m = ?$



$m=2$

$(x_0, y_0) \quad (x_1, y_1) \quad (x_2, y_2) \quad P_2(x) = a_0 + a_1x + a_2x^2$

$$\begin{cases} P_2(x_0) = a_0 + a_1x_0 + a_2x_0^2 = y_0 \\ P_2(x_1) = a_0 + a_1x_1 + a_2x_1^2 = y_1 \\ P_2(x_2) = a_0 + a_1x_2 + a_2x_2^2 = y_2 \end{cases}$$

$$Aa = y \quad a = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & x_0 & x_0^2 \\ 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \end{bmatrix} \quad y = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \end{bmatrix}$$

IN GENERALE

$(x_0, y_0) \quad (x_1, y_1) \quad \dots \quad (x_m, y_m) \quad m+1 \text{ punti}$

$$P_m(x) = a_0 + a_1x + \dots + a_mx^m \quad P_m(x_i) = y_i \quad i = 0, \dots, m$$

$$a = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_m \end{bmatrix}$$

$$Aa = y$$

$$A = \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \dots & x_m^m \end{bmatrix}$$

$$y = \begin{bmatrix} y_0 \\ \vdots \\ y_m \end{bmatrix}$$

$$\begin{cases} P_m(x_0) = a_0 + a_1x_0 + \dots + a_mx_0^m = y_0 \\ P_m(x_1) = a_0 + a_1x_1 + \dots + a_mx_1^m = y_1 \\ \vdots \\ P_m(x_m) = a_0 + a_1x_m + \dots + a_mx_m^m = y_m \end{cases}$$

Si dimostra che
 $\det(A) \neq 0$ se
 x_i distinti

$\Downarrow$
 $\exists!$ soluzione

MINIMI QUADRATI

$n > m+1$ punti

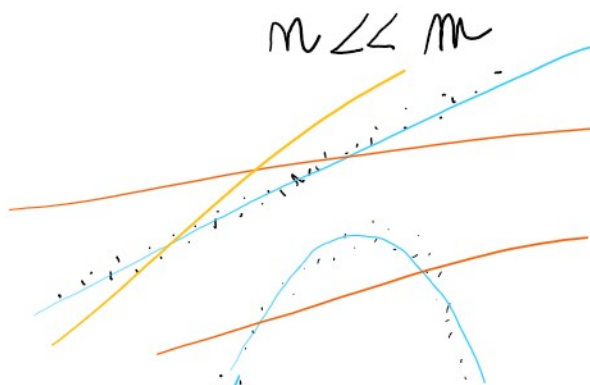
MINIMI QUADRATI

$(x_0, y_0) \dots (x_m, y_m)$

$m+1$ punti

$(m = \text{GRADO DEL POLINOMIO})$

$$m \leq n$$



POLINOMIO DI MINIMI QUADRATI DI GRADO 1 (RETTA DI REGRESSIONE)

$$m=1$$

$(x_0, y_0) (x_1, y_1) \dots (x_m, y_m)$ $m+1$ punti

$$P_1(x) = a_0 + a_1 x \quad a_0 = ? \quad a_1 = ?$$

$$\sum_{i=0}^m (P_1(x_i) - y_i)^2 \text{ MINIMA}$$

$$P(x_i) \\ \downarrow \\ y_i$$

$$S(a_0, a_1) = \sum_{i=0}^m (a_0 + a_1 x_i - y_i)^2$$

minimo di questa funzione di due variabili

$$\begin{cases} \frac{\partial S}{\partial a_0} = \sum_{i=0}^m 2 \cdot (a_0 + a_1 x_i - y_i) \cdot 1 = 0 \\ \frac{\partial S}{\partial a_1} = \sum_{i=0}^m 2 \cdot (a_0 + a_1 x_i - y_i) x_i = 0 \end{cases}$$

$$\begin{cases} \sum_{i=0}^m (a_0 + a_1 x_i - y_i) = 0 \\ \sum_{i=0}^m (a_0 + a_1 x_i - y_i) x_i = 0 \end{cases}$$

$$\begin{aligned} f(x) &= x^3 - 7 \\ \frac{df}{dx} &= f'(x) = 3x^2 \\ f(x, y) &= y^2 x^3 - 7xy + 9y - 7x \\ \frac{\partial f}{\partial x} &= 3y^2 x^2 - 7y - 7 \\ \frac{\partial f}{\partial y} &= 2x^3 y - 7x + 9 \\ f(x) &= (5x - 9)^2 \\ f'(x) &= 2 \cdot (5x - 9) \cdot 5 \end{aligned}$$

$$\sum_{i=0}^m (a_0 x_i + a_1 x_i^2 - y_i x_i) = 0$$

$$\begin{cases} \sum_{i=0}^m a_0 + \sum_{i=0}^m a_1 x_i - \sum_{i=0}^m y_i x_i = 0 \\ \sum_{i=0}^m a_0 x_i + \sum_{i=0}^m a_1 x_i^2 - \sum_{i=0}^m y_i x_i^2 = 0 \end{cases}$$

$$\begin{cases} (m+1)a_0 + \left(\sum_{i=0}^m x_i\right) \cdot a_1 = \sum_{i=0}^m y_i x_i \\ \left(\sum_{i=0}^m x_i\right) a_0 + \left(\sum_{i=0}^m x_i^2\right) a_1 = \sum_{i=0}^m y_i x_i^2 \end{cases}$$

$$A = \begin{bmatrix} \sum_{i=0}^m x_i^0 & \sum_{i=0}^m x_i^1 \\ \sum_{i=0}^m x_i^1 & \sum_{i=0}^m x_i^2 \end{bmatrix}$$

$$b = \begin{bmatrix} \sum_{i=0}^m y_i x_i \\ \sum_{i=0}^m y_i x_i^2 \end{bmatrix}$$

Si dimostra $\det(A) \neq 0$ se a_i distinti

Si dimostra che è
UN PUNTO DI
MINIMO

$\exists!$ la soluzione d.
 $Aa=b$

Polin. di
MINIMI
QUADRATI
di
GRADO m

$$\rightarrow p_m(x) = a_0 + a_1 x + \dots + a_m x^m$$

$$A = \begin{bmatrix} \sum x_i^0 & \sum x_i^1 & \sum x_i^2 & \dots & \sum x_i^m \\ \sum x_i^1 & \sum x_i^2 & \dots & \dots & \sum x_i^{m+1} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \sum x_i^m & \sum x_i^{m+1} & \dots & \dots & \sum x_i^{2m} \end{bmatrix}$$

$$b = \begin{bmatrix} \sum y_i \\ \sum y_i x_i \\ \vdots \\ \sum y_i x_i^m \end{bmatrix}$$

$$a = \begin{bmatrix} a_0 \\ \vdots \\ a_m \end{bmatrix}$$

$$Aa=b$$

polyfit $\rightarrow$ INPUT $X, Y \in \mathbb{R}^{m+1}$

$$p(x) = 2 \cdot (5x - 9) \cdot 5$$

$$\sum_{i=0}^m a_0 = a_0 + a_0 + \dots + a_0$$

$m+1$ volte

$$e = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \quad \sum_{i=0}^m x_i^0$$

$$A \cdot a = b$$

polyfit $\rightarrow$ INPUT $X, Y \in \mathbb{R}^{m+1}$
 m grado polinomio
OUTPUT a

$\rightarrow$ MIN QUAD se $m < M$
INTERPOLAZ. se $m = M$



x



x

$$y = \cos(x)$$

x	y
0	1
π/2	0
π	-1
3π/2	0
2π	1

